

GULI ZHU

Phone: (+1)734-353-8376 Email: gulizhu@umich.edu LinkedIn: guli-zhu-cameron Website: cameron-zgl.github.io

EDUCATION

Beijing Normal-Hong Kong Baptist University (BNBU)	2020.09 – 2024.06
<i>Bachelor of Science in Data Science</i>	<i>Zhuhai, China</i>
University of Michigan – Ann Arbor	2024.08 – 2026.06
<i>Master of Health Data Science</i>	<i>Ann Arbor, MI</i>

RESEARCH EXPERIENCE

★ Model Editing for Medical Multimodal LM (BioMed-AI Lab, University of Michigan)	2025.06 – Present
<i>Graduate Researcher (PI: Prof. Liyue Shen)</i>	<i>Ann Arbor, MI</i>
– Formulated localized factual editing for medical VLMs and designed a six-dimension framework centered on locality vs. generality, modality shift, compositional edits, and temporal stability. – Built a fully reproducible evaluation stack (pre/post-edit scoring, automatic grading, evidence attribution, and audit sampling), enabling ablation-driven analysis rather than anecdotal case studies. – Curated Med-VQA-centric testbeds (VQA-RAD, PathVQA, PMC-VQA, SLAKE) and temporal CT subsets; created stress tests with text/vision perturbations, cross-modality locality probes (edit on CT; probe on CXR/MRI/US), and rare-disease tails. – Benchmarked multiple editing approaches across HuatuoGPT-Vision-34B, LLaVA-Med-7B, and related baselines on our 6-dimension benchmark; preparing a public release of the specification and evaluation code. (<i>Accepted ECCV 2026</i>)	
★ Ultrasound–Diabetes Multimodal Prediction (University of Michigan)	2025.03 – 2026.03
<i>Graduate Researcher (PI: Prof. Liyue Shen)</i>	<i>Ann Arbor, MI</i>
– Built a unified pipeline linking clinical variables (e.g., BMI, age, labs) with skeletal-muscle ultrasound (deltoid, vastus; SAX/LAX) for diabetes/insulin-resistance prediction. – Trained clinical-only baselines and developed segmentation (U-Net/MedSAM) to derive quantitative muscle features (echogenicity/density, thickness, cross-sectional area). – Engineered feature extraction and QC for masks; computed radiomics/texture descriptors; standardized patient-level splits with cross-validated evaluation. – Designed and compared multimodal fusion strategies (early/late fusion) to predict insulin resistance/T2D; documented metrics and ablation findings.	
DIAG Research Subteam – Biomedical Pathology RAG Benchmark (University of Michigan)	2025.01 – 2026.02
<i>Team Lead</i>	<i>Ann Arbor, MI</i>
– Lead subteam under the Multidisciplinary Design Program (MDP) focusing on retrieval-augmented generation (RAG) for biomedical and pathology LLMs. – Designed and implemented a pathology-specific RAG benchmark integrating BioLLM and DeepSeek models with modular retrievers and embeddings. – Developed standardized datasets, retrieval and generation evaluation metrics (e.g., Recall@K, nDCG, faithfulness), and evidence-citation pipelines for benchmarking.	
Final Year Project (FYP): Kimera with SLAM semantic mapping and localization (BNBU)	2023.05 – 2023.12
<i>Student Researcher</i>	<i>Zhuhai, China</i>
– Replicated MIT’s Kimera project and explored semantic segmentation in SLAM. – Investigated impact of semantic segmentation on SLAM localization. – Introduced optimizations to improve project outcomes.	

Large Language Model Evaluation Project (BNBU)	2023.05 – 2023.12
<i>Researcher</i>	<i>Zhuhai, China</i>
– Conducted review of LLM assessment methods; analyzed pros and cons. – Evaluated critical thinking of LLMs with psychology/sociology insights. – Developed methods to assess critical thinking abilities of LLMs.	
Genetic Disease Research Project in Data Science (BNBU)	2022.10 – 2023.04
<i>Project Subteam Leader</i>	<i>Zhuhai, China</i>
– Led data analysis and modeling; trained Decision Trees, Gradient Boosting Trees, Random Forests, and Logistic Regression. – Applied overfitting techniques to enhance performance after identifying suboptimal recall. – Logistic Regression achieved accuracy rate of 93%. – Paper accepted by IEEE MedAI 2023; results implemented in local hospitals.	
Research Assistant for Professor Xuemei Chen (BNBU)	2022.01 – 2022.04
<i>Research Assistant</i>	<i>Zhuhai, China</i>
– Assisted Professor Chen in linguistics research with a focus on data analysis; helped the publication of paper.	
Chinese Academy of Sciences ML Online Internship Program	2021.06 – 2021.09
<i>Intern</i>	<i>Remote</i>
– Constructed a simulation time-series model for Covid-19 using machine learning and data visualization tools.	

INTERNSHIP

ByteDance – Machine Learning Intern	2024.03 – 2024.07
<i>Intern, TikTok Endorsement Project</i>	<i>Shanghai, China</i>
– Developed multimodal recommendation model for TikTok product endorsement. – Built product recognition (CV) and sentiment analysis (NLP). – Designed algorithms for clarity analysis, keyframe extraction, clothing item recognition. – Integrated LLMs to enhance recommendation accuracy and intent analysis.	
UIC Mathematics Summer Course Internship (TA)	2021.06 – 2021.08
<i>Teaching Assistant</i>	<i>Zhuhai, China</i>
– Wrote lesson plans, created courseware, designed course activities.	
Guangzhou Institute of Tropical and Marine Meteorology	2022.06 – 2022.08
<i>Research Intern</i>	<i>Guangzhou, China</i>
– Analyzed typhoon forecasting errors using data-based methods. – Verified results with multiple models and visualization tools.	

PUBLICATION

- Zhu, G.**, Wu, C., & Shen, L. (2026). *Evaluating and Understanding Model Editing for Medical Vision Language Models*. First-author manuscript accepted by ECCV 2026.[<https://arxiv.org/abs/2607.05310>]
- Wang, D., Huang, S., Cao, J., Feng, Z., Jiang, Q., Zhang, W., Chen, J., Liu, C., **Zhu, G.**, Liao, W., Zhang, L., Guo, W., Liu, L., Yang, J., & Qiuping, L. (2023). *A Multivariate Logistic Regression Model with Over Sampling for Bronchopulmonary Dysplasia Associated with Pulmonary Hypertension in Very Preterm Infants: A Multi-center Retrospective Study*. Proceedings of The First IEEE International Conference on Medical Artificial Intelligence (IEEE MedAI 2023).
- Wang, D., Huang, S., Cao, J., Feng, Z., Jiang, Q., Zhang, W., Chen, J., Liu, C., **Zhu, G.**, Liao, W., Zhang, L., Guo, W., Liu, L., Yang, J., & Qiuping, L. (2024). *A comprehensive study on machine learning models combining with oversampling for bronchopulmonary dysplasia-associated pulmonary hypertension in very preterm infants*. Respiratory Research. [Link]

Vu, K., Maddox, A. M., Tonon, C. W., **Zhu, G.**, Wang, T., Opperman, R. M., Ding, Q., Bai, Z., Liu, Z., Wang, Z., Rao, A., & Yoon, D. (2025, June). *BOARD #92: WIP: Generative AI-based Learning Tutor for Biomedical Data Science (GAIL Tutor BDS)*. Proceedings of the 2025 ASEE Annual Conference & Exposition. [Link]

Khan, S., Klochko, C. L., **Zhu, G.**, Ma, D., Wu, C., & Shen, L. (2025). *Early Noninvasive Prediction of Insulin Resistance through Muscle Ultrasound Radiomics*. Radiological Society of North America (RSNA 2025), Scientific Presentations — Oral (accepted), Chicago, IL, Nov–Dec 2025.

OTHER INFORMATION

Institution rename: UIC = BNU–HKBU United International College (now renamed as BNBNU).

Grants & Computing Allocations

- **NSF ACCESS Allocation (CIS261075)**, 2026 — PI. Awarded 8,000 GPU-hours & 20 TB storage on NCSA Delta (est. value \$4,254) for *Modality-aware Model Editing for Reliable Medical Vision-Language Models*. [Link]

Honors

- Outstanding Project Award, 2021 "Jingniu Hui" Cup UIC Innovation and Entrepreneurship Competition
- 2021 UIC Outstanding Volunteer
- Outstanding Volunteer, 7th UIC Fu Chong Youth Entrepreneurship Forum
- Province Second Prize, 2022 National Mathematical Modeling Competition
- Best Poster, UIC Computer Science Exhibition 2023
- Guangdong Medical Valley Scholarship, 2023

Extra-curricular Activities

- Project Manager, Cyber Academy Project team of UIC Enactus (developed 150-hour programming course).
- Undersecretary, Intelligence Capital Department of UIC Enactus.

Computer Skills

Languages: Python, R, SQL, Bash, C

ML/AI: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LoRA, PyTorch Lightning, XGBoost

Data & Distributed: pandas, NumPy, Dask, Spark, Hadoop

MLOps & HPC: Docker, Git, Conda, CUDA, Slurm, Weights & Biases, Linux/Unix

Web/Services: FastAPI, Flask, Django, HTML/CSS